8th

Uniform Distribution

(1) Continuous uniform Distribution (pdf)
(2) Discrete uniform Distribution (pmf)

(1) Continuous uniform Distribution.

Notation :- $U(a, b)$

$-\infty < a < b < \infty$

$$pdf = \begin{cases} \frac{1}{b-a} & x \in [a, b] \\ 0 & \text{otherwise} \end{cases}$$

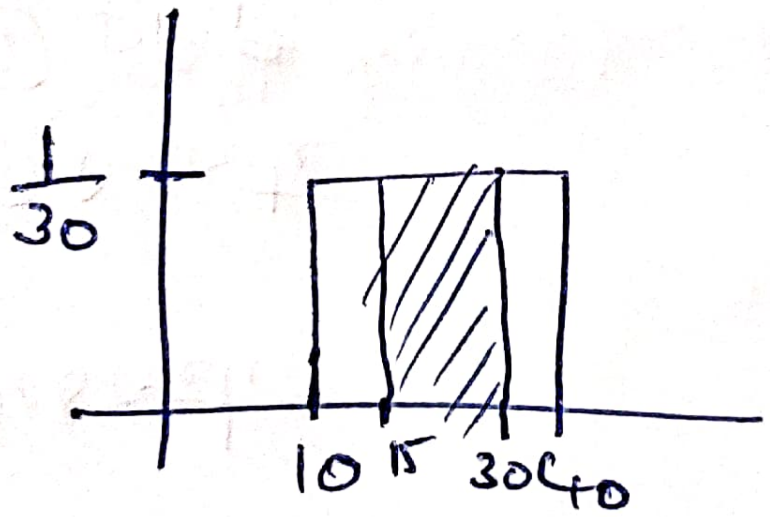
$$Cdf = \begin{cases} 0 & \text{for } x < a \\ \frac{x-a}{b-a} & x \in [a, b] \\ 1 & \text{for } x > b \end{cases}$$

$\text{Mean} = \frac{1}{2}(a+b)$ $\quad$ $\text{Median} = \frac{1}{2}(a+b)$

$\text{varie} = \frac{1}{12}(b-a)^2$

Eg:- No. of Samoso sold at a stall is uniformly distributed with a maxim of 40 Samosa and a min of 10

i) probability of daily sales to fall betw 15 and 30?



$P(15 \leq x \leq 30) = (x_2 - x_1) \times \frac{1}{b-a}$

$= (30-15) \times \frac{1}{30}$

$= \cancel{15} \times \frac{1}{\cancel{30}\,2} \Rightarrow \frac{1}{2} \Rightarrow 0.5$

$pb(x \geq 20)$

$20 \leq x \leq 40) = (40-20) * \frac{1}{30} = 0.66$

$= 76.6\%$

② Descrete

① Descrite

② pmF

Eg :- Rolling a dice $\{1, 2, 3, 4, 5, 6\}$

fair dice a b

$1/n \Rightarrow n = b - a + 1$

Note $U(a, b)$

$b \geq a$

$pmF = 1/n$